

Skills Summary

Reading and Comparing Multi-Digit Whole Numbers

Use this reference while completing your worksheet or when studying.

1. What a digit is worth (place-value-meaning)

Rule: a digit's *value* is the digit times the value of its place.

hundred thousands	ten thousands	thousands	hundreds	tens	ones
100,000	10,000	1,000	100	10	1

Each place is **ten times** the place on its right, so moving one column left multiplies the worth of a digit by 10.

Example: In 82,417, what is the value of the digit 2?

Step 1. The question asks what the 2 is *worth*, so first find which column it is in, then multiply by that column's value.

Step 2. Counting from the right: 7 ones, 1 ten, 4 hundreds, 2 thousands. So the 2 is $2 \times 1,000 = 2,000$.

Try it: In 59,304, what is the value of the digit 9?

check: 9,000

2. Expanded form and word form (expanded-and-word-form)

Three ways to write one number:

standard form 43,206

expanded form $40,000 + 3,000 + 200 + 6$

word form forty-three thousand, two hundred six

Zeros hold places. A place with nothing in it still needs a 0 in standard form, or every digit to its right slides into the wrong column.

Example: Write $40,000 + 3,000 + 200 + 6$ in standard form.

Step 1. Each addend names one place, so write the digits into their columns and fill empty columns with 0.

Step 2. 4 ten thousands, 3 thousands, 2 hundreds, **no** tens (write 0), 6 ones: **43,206**.

Try it: Write $700,000 + 20,000 + 50 + 1$ in standard form.

check: 720,051

3. Comparing and ordering (compare-order)

Rule: more digits means a bigger whole number. With the *same* number of digits, line the numbers up and compare from the **left**; the first place where the digits differ decides it — everything to the right of it is irrelevant.

$<$ means *is less than*, $>$ means *is greater than*. The open side always faces the larger number.

Example: Compare 26,458 and 26,485.

Step 1. Both have five digits, so a digit count decides nothing — scan from the left for the first column where they disagree.

Step 2. Ten thousands $2 = 2$, thousands $6 = 6$, hundreds $4 = 4$, tens 5 against 8: first difference, and $5 < 8$, so $26,458 < 26,485$.

Try it: Write $<$ or $>$: 71,309 ____ 71,093

$<$ check

4. Rounding to estimate (round-estimate)

Rule: to round to a place, look at the digit *one place to the right*.

0–4: keep the rounding digit (round down). 5–9: add one to it (round up).

Every digit to the right of the rounding place becomes 0.

To **estimate a sum**, round each number first, then add the rounded numbers.

Example: Round 63,472 to the nearest thousand.

Step 1. Rounding to thousands, so the thousands digit is the one that may change and the hundreds digit is what decides.

Step 2. Thousands digit 3, hundreds digit 4. Since $4 < 5$, keep the 3 and zero the rest: **63,000**.

Try it: Round 28,715 to the nearest thousand.

29,000 check

(!) Watch out: Rounding is not comparing. 34,650 and 34,509 both round to 35,000, but they are not equal — compare the *original* numbers when a problem asks which is greater.
